

Module/Pathway/Taxon Review: Bacterial Type II Secretion in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu03070` ("Bacterial secretion system" overview map) **Module scope reviewed:** Type II secretion system (T2SS / Xcp / general secretory pathway main terminal branch) **Curation verdict: Module COVERED** — all four canonical steps map to specific KT2440 genes; the supplied candidate list requires substantial revision.

1. Executive Summary

Pseudomonas putida KT2440 encodes a **single, complete Type II secretion system (T2SS)** of the Xcp subfamily in a contiguous chromosomal cluster spanning **PP_1042 and PP_1045–PP_1055**, plus the shared prepilin peptidase **PilD (PP_0632)**. Every step of the generic T2SS module — (1) inner-membrane platform, (2) cytosolic secretion ATPase, (3) pseudopilus assembly, and (4) outer-membrane secretin — maps cleanly to identified genes with **no gaps**. On satisfiability grounds the module should be marked **covered** for this organism.

The principal curation problem is **not** module satisfiability but the **quality of the supplied candidate list**. The 61 candidates inherited from KEGG bucket `ppu03070` are drawn from a broad "Bacterial secretion system" overview map, not from a T2SS-specific module. As a result the list is heavily contaminated: only **10 of 61 candidates** belong to the genuine Xcp T2SS, while the remaining ~50 are members of *other* secretion machineries (Sec/Tat translocons, three Type VI secretion clusters, a Type I/RTX system, a two-partner/Type Vb system) or are over-propagated T2SS-family homologs that actually belong to **Type IV pilus (T4P)** and orphan loci. Simultaneously, the list **omits three genuine Xcp components** (xcpZ/GspM = PP_1054, gspN = PP_1055, and the essential prepilin peptidase pilD = PP_0632).

The actionable recommendations are therefore: (a) mark the T2SS module **covered**; (b) **add** PP_1054, PP_1055, and PP_0632 to the module; (c) **exclude** the ~46 out-of-module genes belonging to Sec/Tat, T6SS, T1SS/RTX, and T5SS/TPS systems (they belong to their own

modules); and (d) flag ~8 "type II secretion"-labeled genes (PP_3423, PP_3424, the PP_3474–3484 locus, PP_5190) as **candidate_uncertain** / **over-annotated** because they are Type IV pilus or orphan homologs that share Pfam domain families with T2SS but lack a cognate secretin+ATPase+pseudopilus operon.

2. Target-Organism Pathway Definition

What the module includes

The Type II secretion system (also called the **general secretory pathway main terminal branch**, **GSP-MTB**, or in *Pseudomonas* the **Xcp** system) secretes **folded** proteins from the **periplasm** across the **outer membrane** into the extracellular milieu. It is a two-step pathway: substrates are first delivered to the periplasm by the Sec or Tat translocons (which are *upstream* of, and not part of, T2SS), and then the T2SS machine translocates the folded substrate across the outer membrane. As established in the foundational review of T2SS architecture, "the multi-gene family necessary for secretion of these enzymes is now known as the type II system or the main terminal branch (MTB) of the general secretion pathway (GSP)" ([PMID: 9641973](#)).

The machine has four functional/structural sub-assemblies, which correspond to the four generic module steps:

Step	Sub-assembly	Function
1	Inner-membrane (IM) platform	Anchors the machine; couples ATPase to pseudopilus (GspC/F/L/M)
2	Secretion ATPase	Cytoplasmic motor energizing the machine (GspE)
3	Pseudopilus	Piston-like fiber that pushes substrate through the secretin (GspG–K major/minor pseudopilins, matured by prepilin peptidase PilD)
4	Outer-membrane secretin	Gated channel in the outer membrane (GspD / XcpQ)

Which neighboring pathways must be kept separate

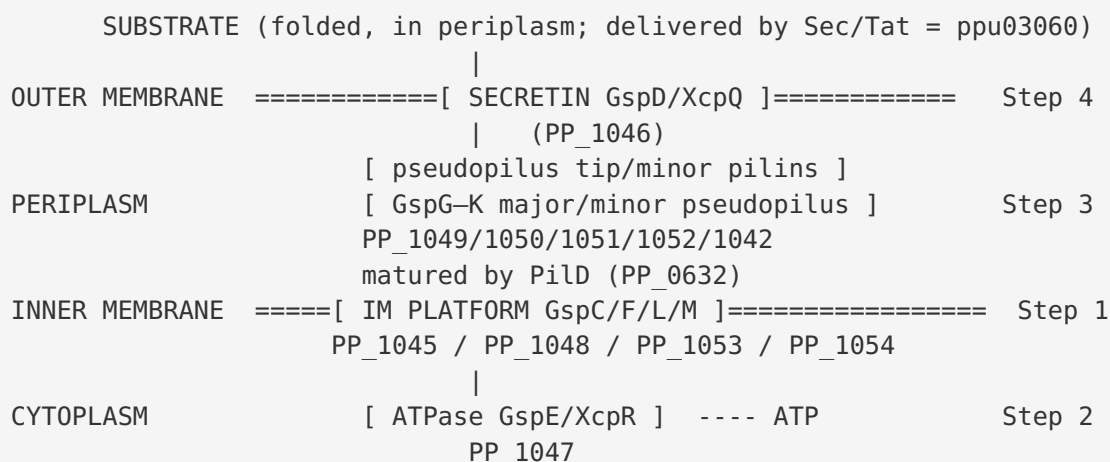
The KEGG [ppu03070](#) overview map lumps **all** protein secretion systems together. For a T2SS-specific module the following must be treated as **separate modules/buckets**:

- **Sec translocon and Tat pathway** (KEGG [ppu03060](#), "Protein export"): SecA/B/D/E/F/G/Y, YajC, YidC, Ffh, FtsY, TatA/B/C. These deliver substrates *to* the periplasm; they are prerequisites for, but not components of, T2SS.
- **Type VI secretion (T6SS)**: contractile injection systems — three separate clusters in KT2440.
- **Type I secretion / RTX (ABC-transporter + MFP + TolC)**: one-step ATP-driven export.
- **Type V / two-partner secretion (TPS, autotransporters)**: Sec-dependent OM insertion.
- **Type IV pilus (T4P) and Tad/Flp appendages**: share the GspD/GspE/GspF/GspG domain families with T2SS but are motility/adhesion machines, not secretion systems.

Alternate names / database definitions

- *Pseudomonas* nomenclature: **Xcp** (extracellular protein) system; genes xcpP–Z + xcpA (=pilD).
 - Generic nomenclature: **Gsp** (general secretory pathway), genes gspC–gspO.
 - Correspondence: XcpP=GspC, XcpQ=GspD, XcpR=GspE, XcpS=GspF, XcpT=GspG, XcpU=GspH, XcpV=GspI, XcpW=GspJ, XcpX=GspK, XcpY=GspL, XcpZ=GspM, and the prepilin peptidase XcpA=PilD.
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3. Expected Step Model and KT2440 Coverage



Module step	Canonical components	KT2440 gene(s)	Status
1. IM platform	GspC, GspF, GspL, GspM	PP_1045 (xcpP/GspC), PP_1048 (gspF), PP_1053 (xcpY/GspL), PP_1054 (xcpZ/GspM)	Covered
2. Secretion ATPase	GspE	PP_1047 (gspE/XcpR)	Covered
3. Pseudopilus	GspG (major) + GspH-K (minor), PilD peptidase	PP_1049 (gspG), PP_1050 (xcpU/GspH), PP_1051 (xcpV/GspI), PP_1052 (xcpW/GspJ), PP_1042 (xcpX/GspK); PilD = PP_0632	Covered
4. OM secretin	GspD	PP_1046 (xcpQ/GspD)	Covered

Accessory component GspN = PP_1055 completes the cluster. The canonical Xcp system "requires at least 12 xcp gene products (XcpA and XcpP to -Z)" ([PMID: 11208795](#)); KT2440 encodes the full XcpP-Z set contiguously plus XcpA/PilD elsewhere, so **all four steps are satisfied**.

4. Candidate Genes and Evidence

4.1 High-confidence T2SS components (the true Xcp cluster)

These 10 candidates (plus 3 additions) constitute the genuine T2SS. Evidence type: UniProt annotation of proteome UP000000556 combined with contiguous operon structure and canonical Xcp/Gsp domain assignments.

Gene	Locus	UniProt	Role in T2SS	Module step	Caveat
xcpX	PP_1042	Q88P11	GspK minor pseudopilin	3	—
xcpP	PP_1045	Q88P08	GspC IM platform / secretin linker	1	—
xcpQ	PP_1046	Q88P07	GspD outer-membrane secretin (584 aa, OM)	4	Diagnostic OM channel
gspE	PP_1047	Q88P06	GspE/XcpR secretion ATPase	2	Energizing motor
gspF	PP_1048	Q88P05	GspF IM platform (polytopic)	1	Domain shared with T4P
gspG	PP_1049	Q88P04	GspG major pseudopilin	3	—
xcpU	PP_1050	Q88P03	GspH minor pseudopilin	3	—
xcpV	PP_1051	Q88P02	GspI minor pseudopilin	3	—
xcpW	PP_1052	Q88P01	GspJ minor pseudopilin	3	—
xcpY	PP_1053	Q88P00	GspL IM platform	1	—
xcpZ	PP_1054	Q88NZ9	GspM IM platform (128 aa)	1	Missing from candidate list — ADD
gspN	PP_1055	Q88NZ8	GspN accessory	(accessory)	Missing from candidate list — ADD
pilD	PP_0632	Q88Q64	Prepilin peptidase (EC 3.4.23.43)	3 (maturation)	Missing; shared with T4P — ADD

Curation-relevant note: The ATPase XcpR/GspE role is directly supported by experimental work in the sister species *P. aeruginosa*: "XcpR, a putative nucleotide-binding protein, is essential for the secretion process across the outer membrane" (PMID: 9882649) and "Assembly of XcpR in the cytoplasmic membrane is required for extracellular protein secretion." Species transfer *P. aeruginosa* → *P. putida* is **strong** here: both are fluorescent pseudomonads with orthologous, syntenic xcp clusters, and cross-species Xcp machinery exchange has been demonstrated (PMID: 11208795).

4.2 Out-of-module candidates (belong to other secretion systems)

These should be **excluded** from the T2SS module and re-filed to their own modules.

Group	KT2440 candidates	Correct module	Why excluded
Sec/Tat translocons	yidC (PP_0006), secE (PP_0441), secY (PP_0474), yajC (PP_0834), secD (PP_0835), secF (PP_0836), tatC-I (PP_1039), tatB-I (PP_1040), tatA-I (PP_1041), secA (PP_1345), ffh (PP_1461), tatA-II (PP_5016), tatB (PP_5017), tatC-II (PP_5018), secB (PP_5053), ftsY (PP_5111), secG (PP_5706)	ppu03060 (Protein export)	Deliver substrates to periplasm; upstream of T2SS, not part of it
Type VI secretion (3 clusters)	vgrG-I/II/III (PP_2614, PP_3386, PP_4049), hcpC-I/II (PP_2615, PP_4082), TssM (PP_2627, PP_4071), TssJ (PP_2618, PP_3094, PP_4079), ClpV (PP_3095), ImcF (PP_3091), DotU (PP_2616, PP_3092, PP_3385), Hcp effectors (PP_3089, PP_4886), OmpA (PP_3090), PP_3106, PP_4081	T6SS module	Contractile injection nanomachine; unrelated to T2SS
Type I / RTX + efflux	cyaB (PP_4927), MFP (PP_4926), OM efflux PP_1798, PP_4923	T1SS / ppu01501	One-step ABC-transporter export
Type Vb / TPS	PP_1449 (CdiA-like TPS), PP_1450 (TPS activation/secretion)	T5SS/TPS module	Two-partner secretion

Rationale confirmed by the T2SS definition: T2SS is the "main terminal branch (MTB) of the general secretion pathway" for folded exoproteins (PMID: 9641973), mechanistically and evolutionarily distinct from Sec/Tat, T1SS, T5SS, and T6SS.

4.3 Over-propagated "T2SS" homologs (Type IV pilus / orphan loci)

These candidates carry "type II secretion" labels but are **not** part of the Xcp T2SS. They should be marked **candidate_uncertain** pending full gene review.

Locus	UniProt	Annotation	Actual assignment	Diagnostic
PP_3423 (xcpT) + PP_3424 (xcpS)	Q88HD7 / Q88HD6	GspG pseudopilin + GspF platform	Orphan T4P- like pair	Isolated pseudopilin+platform with NO cognate secretin or ATPase nearby (flanked by histidine kinases PP_3420/3421, RND-MFP PP_3425)
PP_5190	Q88CJ1	"Type II secretion system protein"	MshE/GspE- family pilus ATPase	Carries MshEN (PF05157) + T2SSE (PF00437); isolated, neighbours are Lrp regulator + glycine-cleavage genes — not a T2SS operon
PP_3474–3484 locus (incl. PP_3475/3476/3477 pseudopilins, PP_3478 secretin, PP_3483 ATPase, PP_3480 PilO)	multiple	Various "type II secretion"	Self- contained second pilus/ secretion machine (T4P/Tad- like)	Has secretin (PP_3478, PF00263+PF03958+STN), single assembly ATPase (PP_3483), PilO alignment protein (PP_3480, PF10741), CsgE (PP_3474) — architecture of a pilus, and lacks a retraction ATPase (PilT is elsewhere at PP_5093)
PP_5238	Q88CE3	Fimbrial protein	Fimbrial/CU pilus	Not a secretion component — mark not_expected

The mechanistic basis for these over-annotations is domain sharing: "one of the core components of the Xcp system is the inner-membrane protein XcpS (GspF), homologues of which can be identified in type II secretion machineries as well as in type IV piliation systems"

([PMID: 17464073](#)). Because GspD/PilQ, GspE-family ATPases, GspF/PilC, and GspG-type pilins are common to both T2SS and T4P, **Pfam domain content alone cannot assign a gene to T2SS** — operon context (presence of a cognate secretin + ATPase + full pseudopilus set) is required.

5. Gaps, Ambiguities, and Likely Over-Annotations

Missing from candidate list (genuine components to ADD)

- **PP_1054 (xcpZ/GspM)** — IM platform, part of step 1.
- **PP_1055 (gspN)** — accessory component completing the cluster.
- **PP_0632 (pilD)** — prepilin peptidase, essential for pseudopilus maturation (step 3). Shared with T4P, which is why it was likely filed separately.

Over-annotations to DOWNGRADE (candidate_uncertain)

- **PP_3423, PP_3424** — orphan GspG/GspF pair; no secretin/ATPase.
- **PP_5190** — isolated MshE/GspE-family pilus ATPase.
- **PP_3475, PP_3476, PP_3477, PP_3478, PP_3483** — components of the independent PP_3474–3484 pilus locus, NOT Xcp.

Not expected in the T2SS module

- **PP_5238** — fimbrial protein (chaperone-usheer pilus).

Key ambiguity: the PP_3474–3484 locus

This locus is a **self-contained, independently regulated minor pilus/secretion machine** distinct from both Xcp T2SS and the main Type IVa pilus. It contains a GspD/PilQ secretin (PP_3478, OM, 621 aa), a single GspE/T2SSE ATPase (PP_3483, 598 aa), three GspG-type pseudopilins (PP_3475/3476/3477), a PilO alignment protein (PP_3480), a CsgE curli-assembly homolog (PP_3474), and an in-cluster response-regulator receiver (PP_3484). Critically it has **no retraction ATPase** (the dedicated Type IVa PilT is at PP_5093), so it is neither the Xcp T2SS nor the canonical twitching-motility T4aP. Its true function (a possible Tad-like or minor pilus/secretion appendage) is an **open question** — resolving it would require a full `fetch-gene` review and ideally experimental characterization. Domain content alone is insufficient because these families are shared with T2SS ([PMID: 17464073](#)).

6. Module and GO-Curation Recommendations

Module step	Status	KT2440 genes
1. Inner-membrane platform	covered	PP_1045, PP_1048, PP_1053, PP_1054
2. Secretion ATPase	covered	PP_1047
3. Pseudopilus (+maturation)	covered	PP_1049, PP_1050, PP_1051, PP_1052, PP_1042; PP_0632 (PilD)
4. Outer-membrane secretin	covered	PP_1046

Overall: module_covered. All four steps map to specific genes with no gaps.

Additional curation actions: 1. **Candidate-list revision:** ADD PP_1054, PP_1055, PP_0632; EXCLUDE the ~46 Sec/Tat, T6SS, T1SS/RTX, and T5SS/TPS genes (re-file to their own modules); DOWNGRADE PP_3423, PP_3424, PP_3475–3478, PP_3483, PP_5190 to candidate_uncertain; mark PP_5238 not_expected. 2. **Module boundary confirmation:** The generic four-step module boundary is **correct** for this organism — no revision needed. The problem is entirely at the candidate-provisioning stage, because the source bucket is KEGG's broad `ppu03070` overview map rather than a T2SS-specific module. 3. **GO annotations:** For the confirmed Xcp genes, GO:0015628 (protein secretion by the type II secretion system) is appropriate. For the T4P/orphan homologs, GO:0015628 is likely **over-propagated**; GO:0043683 (type IV pilus assembly) or a generic pilus term is more accurate. No new GO term requests are needed. 4. **New module document:** Consider a separate module document for the PP_3474–3484 pilus locus to capture that it is a distinct machine, preventing repeated mis-filing into T2SS.

7. Genes to Promote to Full fetch-gene Review

Locus	Reason	Priority
PP_0632 (pilD)	Essential shared prepilin peptidase; confirm single copy serves both Xcp and T4P	High
PP_1046 (xcpQ/GspD)	Diagnostic secretin; confirm OM localization and gating	High
PP_3478 (secretin), PP_3483 (ATPase)	Anchor of the ambiguous PP_3474–3484 locus; determine T4P vs. Tad vs. minor-pilus identity	High
PP_3423 / PP_3424	Orphan GspG/GspF pair — determine whether functional or pseudogenized fragment	Medium
PP_5190	Isolated MshE/GspE ATPase — confirm pilus (not T2SS) assignment	Medium
PP_1054 (xcpZ/GspM), PP_1055 (gspN)	Confirm addition to module and cluster co-transcription	Medium

8. Limitations and Knowledge Gaps

- **Species transfer:** Direct functional/mutant evidence for T2SS in *P. putida* KT2440 specifically is limited; the strongest experimental data (XcpR essentiality, cross-species Xcp exchange) come from *P. aeruginosa* ([PMID: 9882649](#), [PMID: 11208795](#)). Transfer to KT2440 is judged **strong** on the basis of orthology and synteny, but formal KT2440 secretion assays were not located in this review.
- **Substrate identity:** The extracellular substrates secreted by the KT2440 Xcp system were not systematically enumerated here. In *P. aeruginosa*, T2SS secretes lipases/esterases and other exoenzymes; a candidate KT2440 substrate class (e.g., lipolytic enzymes) is plausible by analogy but not confirmed for this strain.
- **Ambiguous locus:** The precise function of PP_3474–3484 (T4P vs. Tad-like vs. minor pilus/secretion) remains unresolved from sequence alone.
- **Assignments are annotation/synteny-based:** Component-to-step mapping rests on UniProt annotation, Pfam/InterPro domains, and operon context, not on KT2440 experimental data.

This is appropriate for curation but should be flagged as homology-inferred, not experimentally verified, for the target strain.

9. Evidence Base

PMID	Title (abbrev.)	How it supports the review
9641973	<i>Macromolecular assembly and secretion across the bacterial cell envelope: type II protein secretion systems</i>	Defines T2SS as the GSP main terminal branch for folded exoproteins — the module boundary used to include Xcp and exclude Sec/Tat, T1SS, T5SS, T6SS
11208795	<i>Exchange of Xcp (Gsp) secretion machineries between <i>P. aeruginosa</i> and <i>P. alcaligenes</i></i>	Establishes the canonical ≥ 12 -component Xcp set (XcpA + XcpP–Z), the completeness template; demonstrates cross-species Xcp machinery transfer
9882649	<i>Assembly of XcpR in the cytoplasmic membrane is required for extracellular protein secretion in <i>P. aeruginosa</i></i>	Direct experimental evidence for the XcpR/GspE ATPase (KT2440 PP_1047) as the essential energizing component (module step 2)
17464073	<i>Interaction domains in the <i>P. aeruginosa</i> type II secretory apparatus component XcpS (GspF)</i>	Documents that GspF/XcpS-type components occur in both T2SS and Type IV pili — the mechanistic basis for over-propagated T2SS annotations on T4P genes (PP_3423/3424, PP_3474–3484)
10559163	<i>A novel lipolytic enzyme located in the outer membrane of <i>P. aeruginosa</i></i>	Context on <i>Pseudomonas</i> surface/secreted lipolytic enzymes and alternative (autotransporter) secretion routes — background for distinguishing T2SS substrates from other export pathways

(The azoreductase paper [PMID: 19238379] surfaced during literature search is not relevant to T2SS and is not used for T2SS claims.)

10. Proposed Follow-up Experiments / Actions

1. **Curation (immediate):** Apply the candidate-list revision — add PP_1054/PP_1055/PP_0632, exclude the ~46 out-of-module genes, downgrade the ~8 over-annotated T4P/orphan homologs, mark PP_5238 not_expected. Mark the module **covered**.
 2. **Full gene reviews:** Promote the six loci in Section 7 to `fetch-gene`, prioritizing PP_0632, PP_1046, and the PP_3478/PP_3483 anchors of the ambiguous locus.
 3. **Experimental (target strain):** A KT2440 xcpQ (PP_1046) or xcpR (PP_1047) deletion coupled with secretome comparison (culture-supernatant proteomics) would provide the first direct KT2440 evidence and identify native substrates.
 4. **Locus characterization:** Transcriptional and mutant analysis of PP_3474–3484 to determine whether it functions as a pilus, adhesion appendage, or minor secretion system, resolving the main open question.
 5. **Provisioning fix:** Replace the KEGG `ppu03070` overview-map bucket with a T2SS-specific ortholog seed (e.g., a curated Xcp/Gsp gene set) to prevent recurrence of the contamination pattern in future automated candidate lists.
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Prepared for manual module satisfiability and gene-annotation curation. Confidence: module coverage HIGH (homology + synteny + strong cross-Pseudomonas experimental support); candidate-list dispositions HIGH for the Xcp core and Sec/Tat/T6SS/T1SS/T5SS exclusions, MEDIUM for the T4P/orphan downgrades pending full gene review.
